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Industrial Internship Report on

"IoT Based Industrial Worker Saftey and Hazard Detection System"

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was (My project was "IoT Based Industrial Worker Safety and Hazard Detection System" used to detect gas leakage, fire, smoke, and high temperature in industries using IoT and Embedded Systems): • Gas leaks • Fire outbreaks • Carbon monoxide poisoning • High temperatures • Structural vibrations These hazards are **not always visible** and can cause serious injury or death if not detected early) This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1 Preface

The Four-week Industrial Internship program organized by upskill Campus (USC) in collaboration with UniConverge Technologies Pvt Ltd (UCT) provided me with valuable industrial exposure and practical learning experience in the field of Embedded Systems and Internet of Things (IoT).

During this internship, I worked on the project titled “IoT Based Industrial Worker Safety and Hazard Detection System”. The objective of this project is to improve industrial worker safety by detecting hazardous conditions such as gas leakage, fire, smoke, and high temperature using ESP32 and IoT technology.

This internship helped me improve my technical knowledge in sensor interfacing, IoT communication, Embedded Systems, and real-time monitoring systems. It also improved my problem-solving and documentation skills.

I would like to sincerely thank upskill Campus (USC), UniConverge Technologies Pvt Ltd (UCT), mentors, trainers, and faculty members for their valuable support and guidance throughout the internship.

Finally, I encourage my juniors and peers to participate in internships and technical projects to gain practical knowledge and industrial exposure for future career development.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (uct Insight)

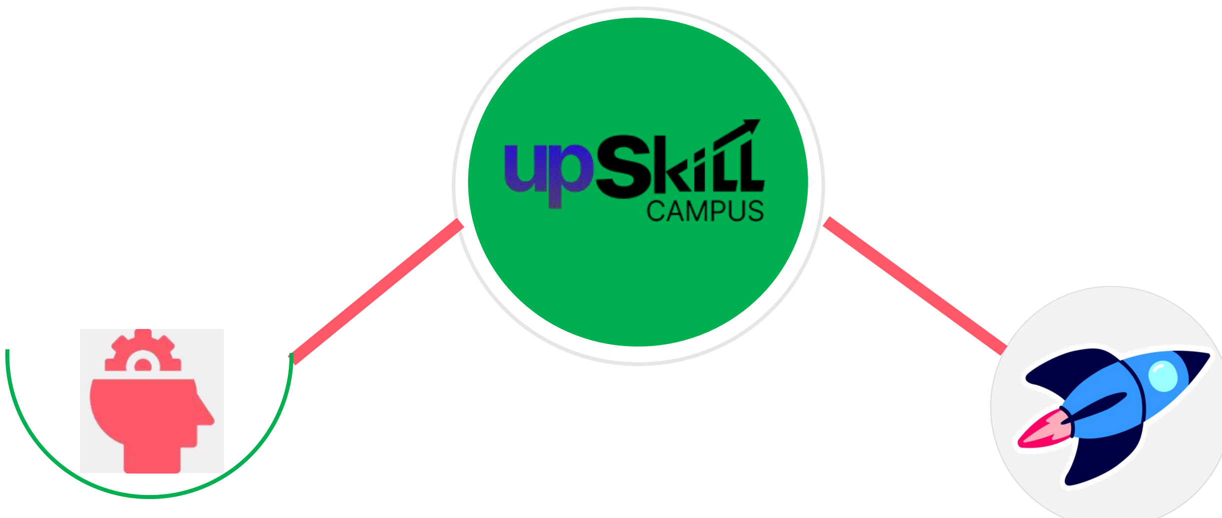
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

2.2 About upskill Campus (USC)

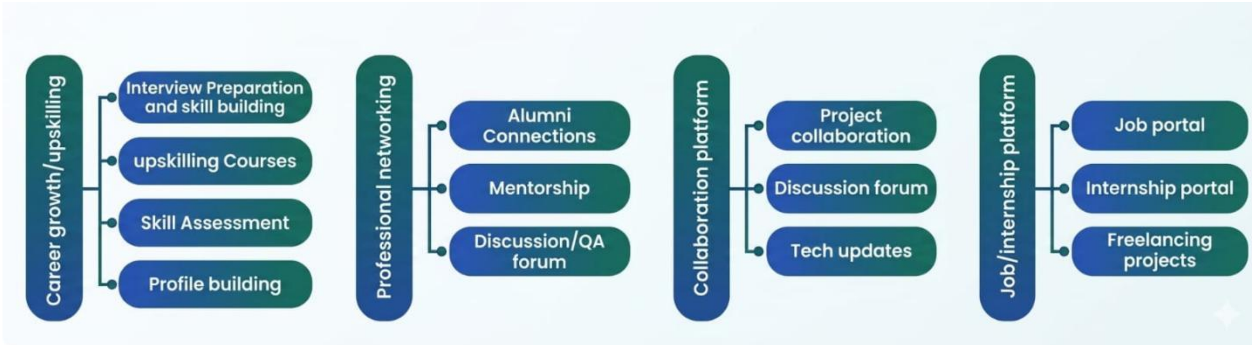
Upskill Campus, in association with **The IoT Academy** and **Uniconverge Technologies**, has facilitated the seamless execution of the complete internship program, providing structured guidance, industry-relevant training, and end-to-end support throughout the internship lifecycle.

Upskill Campus (USC) is a leading career development platform that delivers personalized executive coaching and professional skill enhancement in a highly affordable, scalable, and measurable manner — empowering students and early-career professionals to bridge the gap between academia and industry with confidence.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to:

☛ gain practical knowledge in Embedded Systems and IoT technologies.

☛ understand real-time industrial safety challenges and monitoring systems.

☛ develop problem-solving and technical implementation skills.

☛ improve knowledge in sensor interfacing, cloud monitoring, and ESP32 programming.

☛ enhance communication, teamwork, documentation, and project development skills.

☛ gain industry exposure and improve career opportunities in core engineering domains.

2.5 Reference

[1]ESP32 Microcontroller Datasheet - Espressif System.

[2]MQ2 Gas Sensor Datasheet.

[3]DHT11 Temperature and Humidity Sensor Datasheet.

[4]Embedded System and IoT Concept - Technical Artifacts and Online Resources.

[5]Blynk IoT Platform Documentation

[6]Microcontroller Based Operations - Floyd.

2.6 Glossary

Terms	Acronym
IoT	Internet of Things
MCU	Microcontroller Unit
GPIO	General Purpose Input Output
WiFi	Wireless Fidelity
MQTT	Message Queuing Telemetry Transport

3 Problem Statement

In the assigned Problem Statement

Industrial environments often contain hazardous conditions such as gas leakage, fire, smoke, and abnormal temperature, which can lead to serious accidents, equipment damage, and risk to worker safety. In many industries, traditional monitoring systems mainly depend on manual supervision and basic alarm systems, which may result in delayed detection and slow emergency response.

The lack of real-time monitoring and remote alert systems reduces the overall safety and efficiency of industrial workplaces. Therefore, there is a need for an intelligent and automated safety monitoring system that can continuously monitor industrial conditions and provide immediate alerts during hazardous situations.

To overcome these limitations, the proposed project “IoT Based Industrial Worker Safety and Hazard Detection System” uses Embedded Systems and IoT technology to detect dangerous conditions in real time and improve industrial worker safety through automated monitoring and alert mechanisms.

4 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

Existing industrial safety systems mainly depend on manual monitoring and basic alarm systems for detecting gas leakage, fire, and high temperature. These systems have limitations such as delayed response, lack of real-time monitoring, no remote access, and limited automation features. Due to these issues, industrial accidents and safety risks may increase.

What is your proposed solution?

The proposed system “IoT Based Industrial Worker Safety and Hazard Detection System” is designed using ESP32 and IoT technology to provide real-time industrial safety monitoring. The system continuously monitors gas leakage, fire, smoke, and temperature conditions using multiple sensors. When any abnormal condition is detected, the system automatically activates buzzer alerts and updates the monitoring data through an IoT cloud platform for remote monitoring and quick response.

☛ Real-time industrial hazard monitoring

☛ IoT based remote monitoring system

☛ Faster emergency alert generation

☛ Low-cost and efficient safety solution

☛ Improved worker safety and industrial protection

☛ Smart automation using Embedded Systems and IoT technology

4.1 Code submission (Github link)

https://github.com/Vijayakumar-ece/upskillCampus

4.2 Report submission (Github link) : first make placeholder, copy the link.

https://github.com/Vijayakumar-ece/upskillCampus

5 Proposed Design / Model

The proposed system is designed using ESP32, gas sensors, flame sensors, and temperature sensors to provide real-time industrial safety monitoring. The sensors continuously monitor environmental conditions such as gas leakage, fire, smoke, and abnormal temperature inside industrial areas.

The collected sensor data is sent to the ESP32 microcontroller for processing and analysis. The controller checks the sensor values with predefined safety limits to identify hazardous situations. If any dangerous condition is detected, the system automatically activates buzzer alerts to warn workers immediately.

In addition to local alerts, the system also updates the monitoring status through an IoT cloud platform using WiFi communication. This allows remote monitoring of industrial safety conditions in real time. The proposed system helps improve worker safety, reduce industrial accidents, and provide faster emergency response using Embedded Systems and IoT technology.

5.1 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

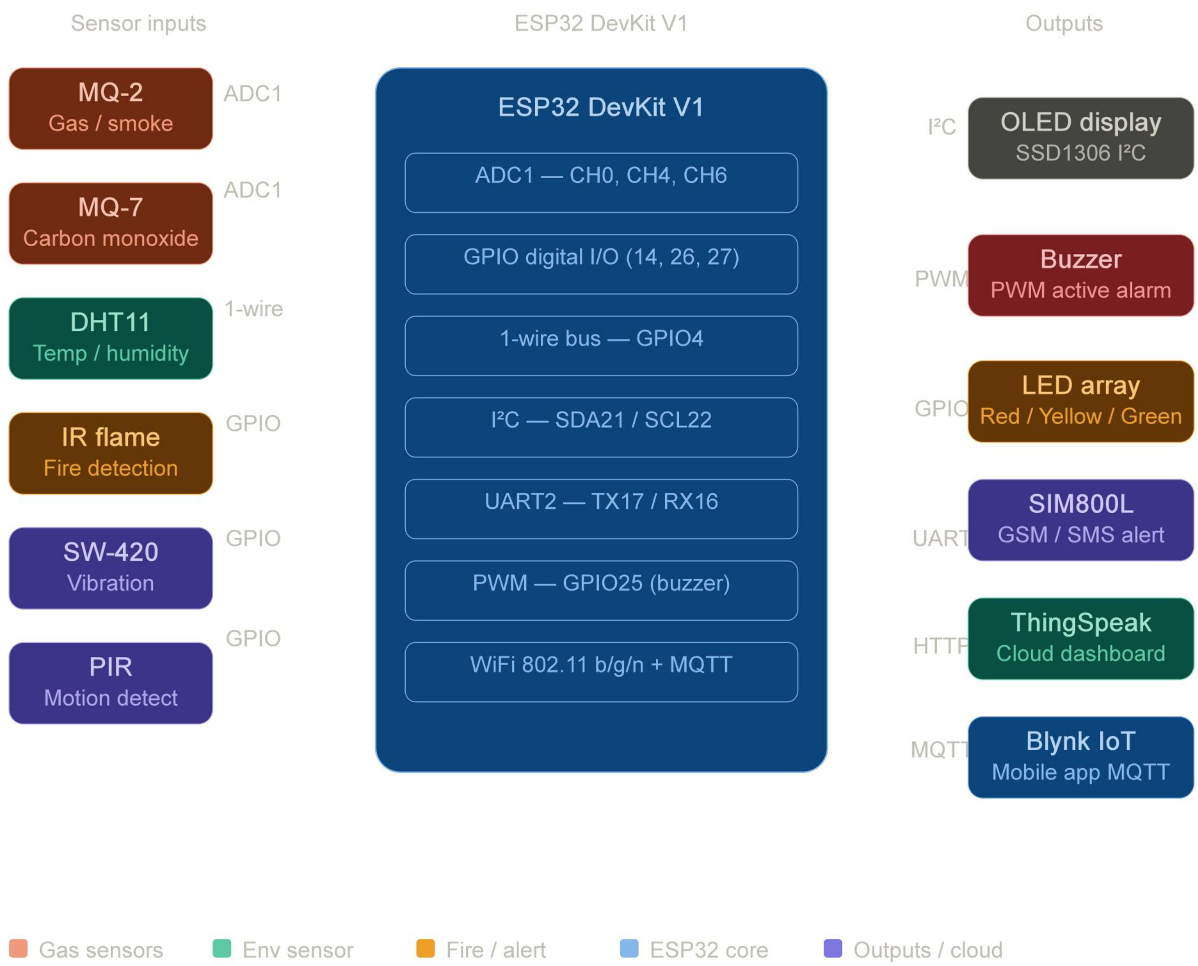


Figure 2: INTERFACE

6 Performance Test

The proposed system was designed to provide real-time industrial safety monitoring with low power consumption and fast response time. The major constraints identified in this project include sensor accuracy, response speed, WiFi connectivity, and continuous monitoring performance.

To overcome these limitations, ESP32 was used for faster processing and stable IoT communication. Multiple sensors were calibrated properly to improve detection accuracy and reliable monitoring. The system was tested under different conditions such as gas leakage, fire detection, and high temperature monitoring.

The test results showed that the system successfully detected hazardous conditions and generated alerts with good response time and stable performance. The proposed solution can be further improved in future by adding advanced sensors, backup power supply, and AI-based predictive monitoring features.

6.1 Test Plan/ Test Cases

The system was tested under different industrial safety conditions such as gas leakage, fire detection, and high temperature monitoring. Test cases were created to verify sensor accuracy, alert generation, and IoT communication performance.

6.2 Test Procedure

The MQ2 gas sensor was tested using smoke and gas exposure. The flame sensor was tested using a controlled flame source, and the temperature sensor was tested under varying temperature conditions. Sensor outputs were monitored through ESP32 and IoT dashboard notifications.

6.3 Performance Outcome

The system successfully detected hazardous conditions and generated real-time alerts through buzzer indications and IoT monitoring platforms. The sensors responded accurately, and the ESP32 provided stable and reliable performance during testing.

7 My learnings

Through this internship program, I gained valuable practical exposure in the field of Embedded Systems and Internet of Things (IoT). Working on the project titled "IoT Based Industrial Worker Safety and Hazard Detection System" helped me understand how real-time industrial problems can be solved using modern embedded and IoT technologies.

During the project development process, I learned ESP32 programming, sensor interfacing, real-time monitoring systems, and loT-based cloud communication. I also gained knowledge about gas sensors, flame sensors, temperature monitoring, and alert generation mechanisms used in industrial safety applications. This internship improved my understanding of hardware and software integration in real-world engineering systems.

In addition to technical skills, I improved my problem-solving ability, analytical thinking, technical documentation, and project planning skills. I also learned how to design a project workflow, identify system limitations, and develop efficient solutions for industrial applications.

The internship provided me with industrial exposure and increased my confidence in handling Embedded and loT-based projects independently. It helped me understand the importance of automation, real-time monitoring, and safety systems in industries. Overall, this internship experience will play an important role in my future career growth and motivate me to further improve my skills in Embedded Systems, IoT, and core electronics engineering domains.

8 Future work scope

Due to time and budget constraints, the following enhancements are proposed for future development of this project:

1. Machine Learning Integration Train a lightweight ML model using TensorFlow Lite on ESP32 to predict hazards before they reach critical levels, reducing false alerts.

2. Multi-Node Mesh Network Deploy multiple ESP32 nodes using ESP-NOW protocol to cover large industrial areas like warehouses and multi-floor plants.

3. GPS-Based Worker Tracking Add a NEO-6M GPS module to locate workers during emergencies and a MAX30100 pulse oximeter to monitor worker health vitals.

4. GSM SMS Alert Fully activate the SIM800L module to send automatic SMS and voice call alerts to supervisors when hazards are detected.

5. Mobile Application Develop a Flutter or MIT App Inventor-based app with Firebase for real-time notifications and historical hazard data access on smartphones.

6. Solar Power Supply Integrate a solar panel with MPPT charge controller and Li-ion battery for self-sustaining operation at remote industrial sites.

7. Cloud Analytics Connect to AWS IoT or Azure IoT Hub for long-term data storage, trend analysis, and automated safety

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